Mazes, Maps, and Memory

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ABSTRACT: *Psychologists studying cognitive processes in animals are particularly interested in the characteristics of memory, the development of cognitive maps, and the use of foraging strategies by predators searching for prey. Laboratory mazes are well suited to study all three of these issues. This article briefly reviews the history of maze testing, focusing on the question of behavioral stereotypy versus behavioral flexibility, and then summarizes the current thinking about memory, maps, and foraging.*

Mazes have played an important role in the development of psychology. At the beginning of this century, when psychology was emerging as a separate discipline, mazes were the apparatus of choice for testing animals and, in many cases, people (see Woodworth, 1938, chap. 6). Consequently, many of the initial descriptions of animal behavior, out of which have developed our current ideas of animal cognition, were developed from observations of animals in mazes. In addition to this empirical contribution, mazes have made conceptual ones as well. Ideas about visual-spatial memory, mental imagery, and reasoning all had developments in the context of maze behavior. Tolman (1959) commented, "The spatial concepts of distance and direction seem to me of the very warp and woof of all our thinking about performances whether these performances involve actual space or mere mechanics or mere logic" (p. 146).

After a brief period of relative disuse, maze testing is enjoying a strong renaissance because it is particularly adapted to addressing the more cognitive types of issues psychologists are investigating today: the structure of memory, the organization of cognitive maps, and the foraging strategies of predators searching for prey. In this article I review some of the highlights in the history of maze testing, discuss current maze procedures and the information they provide about the psychological characteristics of animal cognition, and indicate how the renewed emphasis on spatial behavior can significantly advance our understanding of cognitive processes in animals.

The mazes themselves play an important role in this story. On the one hand, they had a marked influence on the data base that was collected. As will be seen below, structural characteristics of the apparatus suppress some kinds of behaviors and enhance others. Thus the maze itself influenced the types of behaviors observed and the types of theories that were developed from these observations. On the other hand, mazes reflected the theoretical biases of their users. Experimenters had a predilection to address certain types of issues, and the mazes were constructed with these issues in mind. Because of the critical role of the apparatus, diagrams of several of the more important mazes are presented in this article. These diagrams obviously make the experiments clearer, but more importantly, they illustrate the circular interplay among the behaviors observed and the theories proposed, and the manner in which the mazes themselves were both cause and effect in this circle.

In order to provide a maze, the experimenter must define paths and force the animal to remain on these paths. Two general solutions to this problem were developed. In one case, which I call an *alley maze,* the parts were delineated by walls, usually of opaque material. The animal was forced to stay in the alleys formed by these walls either by making the walls high enough to prevent the animal from jumping out or by providing a cover over the top. In the second case, which I call an *elevated maze,* paths were delineated by elevated runways that had at most only a slight edge (about 2 cm high) to provide a foot-

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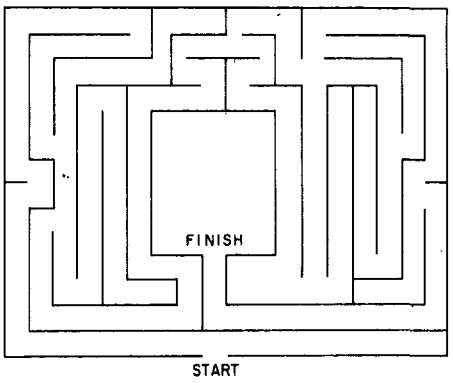


Figure 1. Modified version of the Hampton Court maze—The rat began at the start and was rewarded with food for reaching the finish. (From "An Experimental Study of the Mental Processes of the Rat" by W. S. Small, *American Journal of Psychology*, 1900, *11*, 133–165. Copyright 1900 by Stanley Hall. In the public domain.)

hold for the rat. With the runways placed far enough apart, the animals were discouraged from jumping across the gap and were forced to follow the paths. In both mazes, an incorrect choice led to a cul-de-sac. In the alley maze, this dead end was signified by a wall; in the elevated maze, by the end of an arm and open space.

The Kinesthetic Machine

The current emphasis on behavioral flexibility and cognitive processes stands in marked contrast with the early emphasis on behavioral stereotypy and peripheral mechanisms. The magnitude of this change can best be appreciated by briefly examining the older viewpoint. The progenitor of the early laboratory mazes was constructed following the design of the maze in Hampton Court Gardens, England (Scott, 1931). The living original was wedge shaped, producing many acute and obtuse angles. The laboratory versions were modified so that these angles were 90°, but the general topological organization of the paths through the maze was unaltered. As is indicated in Figure 1, the maze required the animal to find the correct path from the starting point at the outside to the central area in which food was placed. The maze was usually an alley maze, with wooden or metal walls about 10 cm high, but there were occasional exceptions (see, e.g., Small, 1901; Vincent, 1915). The animals most often tested in this maze were rats, but experiments included monkeys, frogs, crawfish, perch, guinea pigs, sparrows, and pigeons (see Watson, 1907, for a review).

Small (1901) introduced the modified Hampton Court maze and trained both domestic and wild rodents (*Mus decumanus*) to run to the center for food. His summary statement included almost all the critical features of animal behavior in mazes:

> In appreciating the results of this series of experiments, about the same facts come into view, only more distinctly, as in the case of the wild gray rats; the initial indefiniteness of movement and the fortuitousness of success; the just observable profit from the first experiences; the gradually increasing certainty of knowledge indicated by increase of speed and definiteness; and the recognition of critical points indicated by hesitation and indecision; the lack of imitation and the improbability of following by scent; the outbreak of the instincts of play and curiosity after the edge of the appetite is dulled. In addition are to be noted the further observations upon the contrast between the slow and cautious entrance into, and the rapid exit from the blind alleys, after the first few trials; the appearance of disgust on reaching the end of a blind alley; the clear indication of centrally excited sensations (images) of some kind; memory (as I have used the term); the persistence of certain errors, and the almost automatic character of the movements in the later experiments. (Small, 1901, p. 218)

Although Small's report included many important observations, the major focus of subsequent experiments was "the almost automatic character of the movements" after the animal had learned the task. Some appreciation of this behavior can be obtained by realizing that the path through one version of the Hampton Court maze was 40 feet (12 m) and was covered by the rats in about 20 seconds, giving an average rate of travel of 2 ft./sec (.6 m/sec; Watson, 1907). Indeed, after extensive training in this type of procedure, rats generally look like a piece of well-oiled machinery, moving smoothly through the maze with no hesitation or jerkiness, rounding off all corners, banking off the centrifugal walls, and generally moving at full tilt throughout.

The automaticity of well-trained rats became almost legendary. In one experiment, rats were trained to run an alley maze following a series of left-hand turns. Then one of the left-hand arms was blocked and the opposite arm was made correct. The rats seemed to be completely oblivious of this change, and "every one of the rats banged his nose into the end of the blind alley with considerable violence" (Gingerelli, 1929, p. 255). In another experiment, hungry rats were trained to run down a straight arm to obtain a small piece of food at the end. Then a large pile of food was placed in the middle of the runway. All rats ran directly over the large pile of food for a number of trials, and about half of them took more than

20 trials before they stopped and ate the large pile of food (Stoltz & Lott, 1964). In a third experiment, rats were trained on an elevated maze to follow a particular pattern of left and right choices. The direction of the correct choices was then reversed so that previously correct arms led to empty space. Here again the rats were relatively insensitive to the changes and "many times . . . went over the ends of the blinds, just catching themselves with their hind claws" (Dennis, 1932, p. 394). In short, by appropriate training, animals in mazes can be turned into apparently unthinking, inflexible machines (cf. Carr & Watson, 1908) that appear to have little regard for the physical or motivational consequences of their actions (see also Fentress, 1976).

For a variety of reasons, early experimenters emphasized the role of kinesthetic sensations and muscular movements in maze performance. In most cases, the reasoning was by the process of elimination, a risky logic even at best and a deficiency that the users of that logic readily admitted (Watson, 1907). The supporting experiments generally attempted to interfere with the ability of the animal to use other sensory modalities (vision, audition, olfaction) by altering either stimuli in the environment or receptors on the animal (Small, 1901; Watson, 1907). Since performance was affected little if at all by these manipulations, the conclusion was drawn that kinesthetic stimuli and muscle contractions must underlie spatial behavior.

One of the most succinct summaries of the "kinesthetic-machine" hypothesis (Fields, 1928, p. 143) was made by one of its opponents:

> The general assumption underlying the function of kinesthesis in learning is that stimuli arise is some sort of temporal sequence from the activity of the muscles as the animal traverses the maze pattern. The further assumption which must be made is that during the process of learning this temporal sequence becomes fixed. . . . The "movement to come" is released at the proper time by the afferent "kinesthetic" impulses aroused by the movement which has just been made. (Dorcus & Gray, 1932, p. 447)

Attacks on the Kinesthetic-Machine Hypothesis

The kinesthetic-machine hypothesis was attacked almost immediately. One of its components, the notion that kinesthesis provided the necessary stimuli for successful spatial discrimination, died an early death. It was refuted by pure logic: "The recognition of the functional value of the kinesthetic sensations and the dominant part they play in such a bit of learning as this has been of the greatest possible value. But the question is immediately raised, however, of what intrinsic value are eyes, ears, etc., if not for learning?" (Vincent, 1915, p. 140). It was refuted by experimentation at both the physiological (Lashley & Ball, 1929) and the behavioral (Gingerelli, 1929) level. Indeed, "the extreme peripheralism and muscle twitch-ism of Watson" (Tolman, 1959, p. 94) was probably taken seriously only by a few of the people studying spatial behavior, even at the time it was proposed.

The second component of the hypothesis, which emphasized the relative inflexibility of behavior and the tendency of animals to repeatedly follow some kind of sequential path through the maze, had a much more profound influence on thinking about behavior and continues to be put forth by the uninitiated as the most likely explanation of spatial behavior. But this component, too, has little support in data. Because of the prevalence of this view, however, the evidence against it needs to be considered in detail.

Alternation and Exploration

In contrast with the early experiments which emphasized the stereotypy of behavior, subsequent experiments emphasized its variability and demonstrated that in many situations animals chose not to repeat their previous choices. One of the simplest examples of this type of behavior is *spontaneous alternation,* usually studied in a T or Y maze (see reviews by Dember & Fowler, 1958; Douglas, 1966). The animal is given two successive trials, usually with an intertrial interval of only a few seconds. The question of interest is whether the arm chosen on the second trial is the same as the one chosen on the first trial. In most experiments, rats choose the opposite arm on about 80% of the tests, a level of alternation characteristic of many animals in a two-choice procedure.

An elegant example of response variability was provided by Dashiell (1930). As is indicated in Figure 2, the maze was an open arena divided into alleys by blocks placed in the middle, making 20 different routes that were equally efficient in terms of the distance required to go from the start to the goal. Rats learned to choose these routes in preference to others, and both the time and the distance required to go from the start to

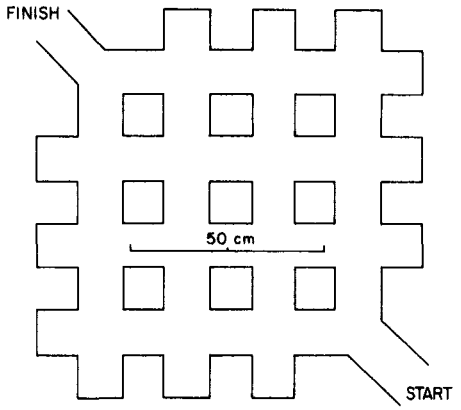


Figure 2. Dashiell's alley maze, which examined the variability of choice behavior—There are 20 different routes from the start to the finish that cover an equivalent distance. (From "Direction Orientation in Maze Running by the White Rat" by J. F. Dashiell, *Comparative Psychology Monographs*, 1930, *7*, No. 32. Copyright 1930 by Comparative Psychology Monographs, Inc. In the public domain.)

the goal decreased markedly in the first few trials. But after the maze had been mastered, rats preferred not to repeat the paths of their previous runs. One rat, for example, used 13 different paths in 21 trials and, during that time, made only three wrong turns.

The marked tendency of rats to explore their spatial environments in exhaustive detail has been well documented ever since the observations of Small (1901), but the efficiency with which this exploration is carried out has generally not been recognized. Bättig and his colleagues have addressed this issue with an ingeniously constructed hexagonal maze (Bättig, Driscoll, Schlatter, & Uster, 1976; Bättig & Schlatter, in press; Uster, Bättig, & Nägeli, 1976; Uster, Note 1). As is indicated in Figure 3, their maze was composed of three concentric alleys interconnected by radiating alleys. Within each section of the maze were photocells connected on line to a computer to monitor the location of the rat in the apparatus. The maze was constructed so that barriers could be placed in the alleys, preventing travel through them; in this way many different configurations could be established using the same basic structure.

Rats were generally given one 6-min. test (with no food in the maze) each day for a number of consecutive days. As would be expected, the rats moved around a great deal during this period, exploring the apparatus. At first their movements were slow and they often repeated sections they had covered previously. But within a few tests, their speed of movement increased and they rarely repeated one section until they had visited all the remaining ones once. Some repetitions, of course, were required to get to the last sections of a dead end, but by following various routes through the apparatus the rat could visit each section once without having to repeat many sections on the way. At the end of about 10 tests, each rat's exploration was highly efficient; all maze sections were visited with a minimal number of repetitions and in a relatively short period of time. (Bättig Note 2) comments:

> The rats . . . patrol the maze progressively by omitting repetitive entries into the same section which becomes evident by the decrease of the number of all movements until each section has been visited once. This improvement of patrolling is dependent upon the sequence and complexity of the presented maze configurations and can also be transferred from one maze configuration to another.

There are several important implications of these results. First, in order for the rats not to have repeated sections of the maze, they must have remembered where they had been. These data indicate that the rats' memory of where they have been is very good, a point I elaborate below.

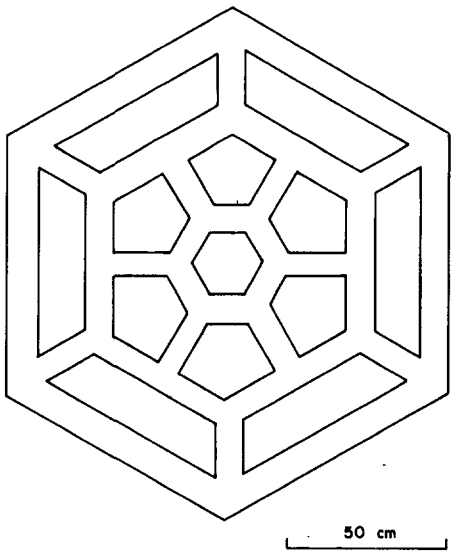


Figure 3. Bättig's enclosed alley maze—Blocks can be placed in the alleys at different points to change the configuration. Photocells throughout monitor the rat's behavior. There is no food reward. (From "Effects of Maze Geometry and Experience on Exploratory Behavior in the Rat" by H. J. Uster, K. Bättig, and H. H. Nägeli, *Animal Learning & Behavior*, 1976, *4*, 84–88. Copyright 1976 by the Psychonomic Society, Inc. Reprinted by permission.)

Second, in order to be efficient in traversing the maze, the rats must have organized their paths so that they didn't place themselves in situations where they would be forced to make repetitions. These data suggest that the rats had formed some kind of a spatial representation of the environment and had planned their behavior appropriately before starting on their path.

Another example of exploration was provided by Barnett and his colleagues, using a *residential maze,* illustrated in Figure 4 (Barnett & Cowan, 1976; Barnett, Dickson, Marples, & Radha, 1978). Unlike other experiments, in which animals live in a home cage and are in the maze only for the test period, these experiments kept rats in the apparatus 24 hours a day. The nest box was at the intersection of the four arms. Each of the four arms differed in terms of the material provided to the resident rat; for example, one arm might contain water; a second, ground corn; a third, wheat; and a fourth might just provide the opportunity to explore.

The question of interest was the manner in which the rat moved among the arms when allowed access to them. The patterns varied, but a very prevalent and distinctive one was a patrol of the environment, characterized by a brief visit (and probably no consummatory behavior) to each of the arms. This patrol might be repeated several times before the rat settled down to some particular activity such as eating, drinking, or grooming. The results of these experiments demonstrated that rats have a systematic pattern of exploration even in very familiar environments.

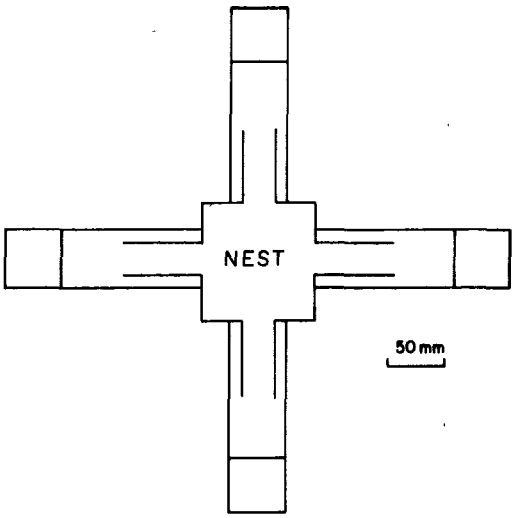


Figure 4. Barnett's residential maze—The animal lives in a nest box in the center. This scale is for mice; a larger scale (75 cm) and a slightly different design is used for rats. (From "The Movements of Wild and Domestic House Mice in an Artificial Environment" by S. A. Barnett and J. L. Smart, *Behavioral Biology,* 1975, *15,* 85–93. Copyright 1975 by Springer Verlag. Reprinted by permission.)

Characteristics of Spatial Memory

Working memory (Honig, 1978), also called short-term memory, is useful for retaining information required for only a short period of time. The memory during the time interval ought to be as accurate as possible, but after the interval is over (and the event for which the memory was required has passed) the memory should be eliminated. A simple example of the use of working memory is dialing a telephone number. While you are dialing, the number ought to be present in working memory, but after you have completed the call this number ought to be removed from working memory (although it might be stored somewhere else), especially if you are about to make another call to a different number.

Working memory in animals has typically been studied with a delayed matching-to-sample (DMTS) procedure. Here a sample stimulus is presented and then removed. After a period of time, the sample stimulus and another stimulus are presented. The animal is rewarded for choosing one from this pair and, in order to respond correctly, must remember which of the stimuli was presented as the sample. Considerable information has been obtained from DMTS procedures about variables such as the duration of the sample stimulus, the length of the delay interval, and the events that happen within the delay interval (D'Amato, 1973; Riley & Roitblat, 1978). But one of the limitations of DMTS as used in operant procedures has been the difficulty of obtaining accurate performance when there is more than one stimulus to be remembered and the delay interval is more than a few seconds. There have been occasional reports of accurate performance in operant procedures with more stimuli (with dolphins; Thompson & Herman, 1977) and longer delays (with monkeys; D'Amato, 1973), but these are rare and have been found only in well-practiced subjects with a long training history. This is in distinct contrast with the high level of performance in a spatial procedure (Tinklepaugh, 1932).

Recently, a spatial analogue of the operant

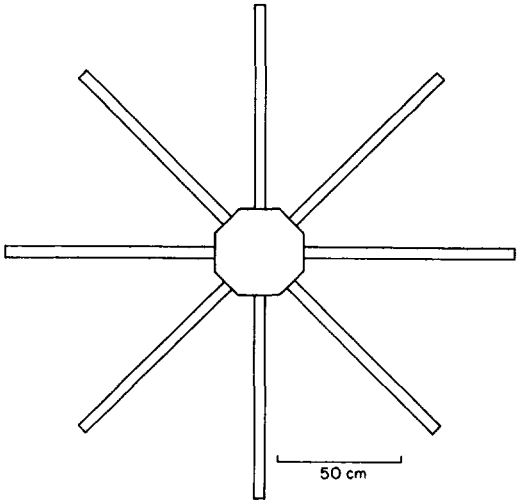


Figure 5. Olton's elevated radial-arm maze—One pellet of food is placed at the end of each arm, and the rat is rewarded for choosing each arm once. (From "Remembrance of Places Passed: Spatial Memory in Rats" by David S. Olton and Robert J. Samuelson, *Journal of Experimental Psychology: Animal Behavior Processes,* 1976, *2,* 97–116. Copyright 1976 by the American Psychological Association, Inc. Reprinted by permission.)

DMTS procedure has been developed. This analogue is useful for two reasons. First, of course, it provides systematic information about the characteristics of working memory in spatial tasks. Second, it is a procedure in which animals are able to remember long lists of items for a considerable period of time. This latter point it important because many of the ideas about memory, from the days of Ebbinghaus (1885) to the present (Anderson & Bower, 1973), have been based on procedures in which the individual is asked to remember lists of items. Consequently, a DMTS procedure that provides an opportunity to obtain information about lists of items is useful for addressing questions about animal memory in a manner that has not been possible before.

The apparatus is a *radial-arm maze,* illustrated in Figure 5. Each of the arms leads away from a center platform like spokes on a wheel. At the beginning of each test, one food pellet is placed out of sight at the end of each arm. A food-deprived animal is placed in the middle of the apparatus and is allowed to choose among the arms. The optimal strategy, of course, is to choose each arm once without repeating any arm.

The original experiments with this maze (Olton & Samuelson, 1976) demonstrated that rats did not use intramaze cues (food cues, odor trails, etc.) or response chains to solve the task. This finding was expected in light of the results from the early maze experiments and has been replicated with even better controls (Olton & Collison, in press). The importance of extramaze visual cues has been demonstrated by Black and his colleagues (Black, Augerinos, & Suzuki, Note 3; Suzuki, Augerinos, & Black, Note 4) in this task and by O'Keefe and Conway (1978) in a related elevated-maze procedure. These data demonstrate that the location of each arm is specified by extramaze cues, many of which may be visual.

Rats perform very well in this task, a finding that is expected for a variety of reasons (see Olton's, 1978, review). On an 8-arm maze, after about 40 tests, they performed almost perfectly, choosing a mean of about 7.9 arms in the first 8 choices (Olton & Samuelson, 1976). On a 17-arm maze, after about 20 tests, they reached asymptote with a mean of about 15 different arms in the first 17 choices (Olton, Collison, & Werz, 1977). These data, in conjunction with those described above, indicate that the rats must be forming a list of places (identified on the basis of extramaze cues) they have chosen that should not be chosen again.

A model of this process has been developed and experiments have been conducted to test this model (see Olton's, 1978, review). The results of these experiments lead to the following conclusions: (a) The capacity of working memory is substantial and can probably accommodate up to about 12 items.[1] (b) Interference takes place among the items in working memory so that as the number of items increases, the accuracy of recall decreases. (c) Decay of information over time does not occur, or if it does, it must become operative only beyond the 20-min. intervals that have been tested so far. (d) Items are retained in working memory irrespective of the serial order of their entry; there was no evidence for either a primacy or a recency effect.[2] (e) Working memory can be reset at the end of a test; the inter-

[1] J. Kubie at the University of Pennsylvania has carried out a mathematical analysis that indicates that a mean of 15 correct responses in the first 17 choices reflects a memory capacity of about 12 items.

[2] Since these experiments, Roberts (Note 5; Roberts & Stephenson, Note 6) has developed a better procedure for testing serial order effects, using a similar apparatus. He has demonstrated a substantial recency effect but no primary effect.

ference among items that had been building up within a test can be reduced before the next test, even when the interval between tests is about the same as the interval between choices within a test. (f) The selection of a new choice appears to take place by a random process; there is no indication of a predictable order for choosing arms, at least when a confinement procedure is used to interrupt ballistic response tendencies.

In summary, the experiments using the radial-arm maze have provided a means for systematic investigation of working memory in animals. A number of questions still remain, of course. Some of them are empirical and have to do with the extent to which animals can perform as well in nonspatial analogues, whether resetting can be placed under discriminative control, and so forth. Other questions are more conceptual and have to do with the distinction between lists of items and maps of places (Black, Nadel, & O'Keefe, 1977; Black et al., Note 3; Suzuki et al., Note 4), the mechanism by which information is eliminated from working memory and the fate of this information, whether or not rats can develop strategies to "chunk" information and improve performance, and so on. In short, these experiments make two contributions: First, they provide us with some specific information about the characteristics of working memory in animals. Second, and more important, they sugggest that through spatially organized behaviors we will be able to obtain answers to questions about animal cognition that have remained relatively intractable in nonspatial procedures.

Cognitive Maps

As indicated previously, the early maze experiments stressed behavioral stereotypy. In contrast, the later experiments, especially those of Tolman, emphasized behavioral flexibility. This was particularly true in those experiments designed around the notion of a *cognitive map*. The characteristics of a cognitive map were never well developed by Tolman, either in his experiments (Tolman, Ritchie, & Kalish, 1946) or in his major theoretical works (Tolman, 1948, 1959). Commonly, one implication of a map is the prediction that an animal ought to be able to take a shortcut if one is available, or take a detour if the original path is blocked. This approach to cognitive maps has had a long history in animal behavior (see Menzel, 1978), and in Tolman's hands it became manifest as the well-known *sunburst maze* (Tolman et al., 1946), illustrated in Figure 6.

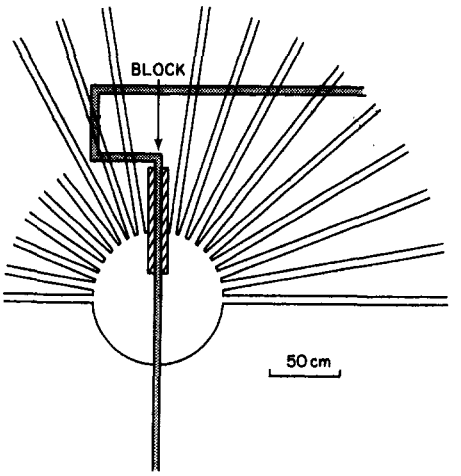


Figure 6. Tolman's sunburst maze—Rats were first trained to run along the shaded path from the start to the goal. Then the path was blocked as indicated, and the rat was allowed to choose among the other arms. (From "Studies in Spatial Learning: I. Orientation and the Short-cut" by E. C. Tolman, B. G. Ritchie, and D. Kalish, *Journal of Experimental Psychology,* 1946, *36,* 13–24. Copyright 1946 by the American Psychological Association, Inc. Reprinted by permission.)

The maze, and the experiment, came in two parts. First, rats were trained to run out the path, indicated by shading in the figure. At this stage, there was no discrimination problem; the rat just followed the single path from the start to the goal. After this training was completed, the original path was blocked and the choice point with the other arms was added. Tolman predicted that the rats would choose the arm oriented in the direction of the goal, indicating that they understood the spatial relation between the start and the goal and could take the shortest path to get there. Indeed, a substantial proportion of rats chose arms oriented toward the goal, even on the first opportunity.

Although this experiment is often cited as a classic in support of cognitive mapping abilities in rats, such a choice is unfortunate. First, the results could not be replicated (Gentry, Brown, & Kaplan, 1947; Gentry, Brown, & Lee, 1948). Second, there was also a flaw in the logic. True, one alternative for the rats was to choose an arm oriented directly toward the goal, providing they had reason to believe that this arm actually went to the goal. But all their prior experience was on an arm that had three right-angle bends in it so

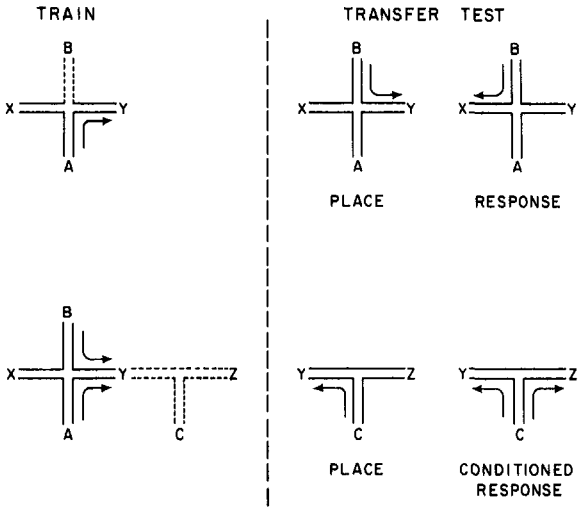


Figure 7. A consideration of the importance of transfer tests in examining maze strategies—Solid lines indicate the portions of the maze used in training; dotted lines indicate portions of the maze not used until the transfer test. Arrows indicate the movement of the rats. For further explanation, see Footnote 3.

that in order to get to the goal they had to begin in a direction away from the goal. An equally intelligent solution when the original path was blocked would be to choose an arm adjacent to it, hoping to intersect the original path beyond the point of the block. Thus, predicting the "correct" choice under these circumstances is very difficult. But Tolman's sunburst experiment contains the core elements necessary for demonstrating shortcut capabilities. First, animals must be made familiar with their environment and trained so that there is a correct response that is predominant. Second, a transfer test must be used to determine whether the animals have learned the relevant spatial relationships. As in Tolman's experiment, this transfer test must be one in which the correct path involves a route never taken previously.[3]

Perhaps because the weaknesses in Tolman's experiment were generally unrecognized, or just because the interests of animal psychologists were changing at the time, there were no other reports of experiments using the necessary transfer procedure to examine the shortcut abilities of rats in mazes. Walker and Olton (Note 7) recently designed an experiment to address this issue. The maze was an elevated interconnected grid of start boxes (A, B, C, and D) and goal boxes, illustrated in Figure 8. The rat was first trained to go from three of the start boxes to the center goal box in order to obtain food. After learning this task, each rat was taken to the fourth start box for a transfer test. The rats had never been in this particular part of the maze before and had never traversed the correct route from the next start box to the goal. The particular set of start boxes used and the order in which they were presented were varied from rat to rat to control for any position bias that might have been present.

Even on the first transfer test, about 80% of the rats made the correct choice, and their performance level improved slightly with further training. Thus, the rats had not learned three

[3] The importance of transfer tests in this type of analysis became particularly clear during the place-versus-response controversy (see Restle's, 1957, review). The question was a simple one. A rat is trained in a T maze to go to the right-hand arm. Several different strategies are possible. One is an egocentric one, based on the relation between the arms and the animal's body position. This strategy was called a *response strategy,* indicating that the animal might have learned to turn right at the choice point. A second strategy is an allocentric one, based on cues not associated with the animal. This strategy was called a *place strategy,* indicating that the animal might have learned to go to the place in which the right-hand arm was located.

One means of distinguishing between these two strategies was to use a transfer test in a *plus maze,* which had two start arms (opposite each other). The rat was first trained to go from one start arm to one goal arm. Then the transfer test took place with the rat starting from the opposite start arm. If the original test had been learned with a response strategy, the rat ought to make the same turn on the transfer test and thus end up in a different arm. If, on the other hand, the original test had been learned with a place strategy, then the rat ought to go to the same arm on the transfer test and make a different turn. These procedures are illustrated in the top portion of Figure 7, which indicates the two mutually exclusive predictions that arise from the response hypothesis and the place hypothesis.

Does it follow that a rat can be trained in a plus maze to follow a place strategy by rewarding the rat for going to the same goal irrespective of the start arm? Although such a conclusion may seem valid at first glance, further consideration indicates that it is inappropriate. True, the animal may have learned a place strategy: Go to this place irrespective of the starting point. But an equally valid alternative is that the animal has learned two conditional response strategies: When beginning a trial from one start arm, turn to the right; when beginning a trial from the other start arm, turn to the left (Blodgett & McCutchan, 1947; Ritchie, 1947; Ritchie, Aeschliman, & Pierce, 1950). The only way to distinguish between these two alternatives is with another transfer test, indicated in the bottom half of Figure 7. If the rat has learned a place discrimination, it ought to go to Point Y on the transfer test. If it has learned a conditional response discrimination, it ought to perform randomly. The important point, then, is that distinguishing among different strategies of solving a task cannot be addressed either by a piece of apparatus or by a method of training. It can be addressed only by a transfer test.

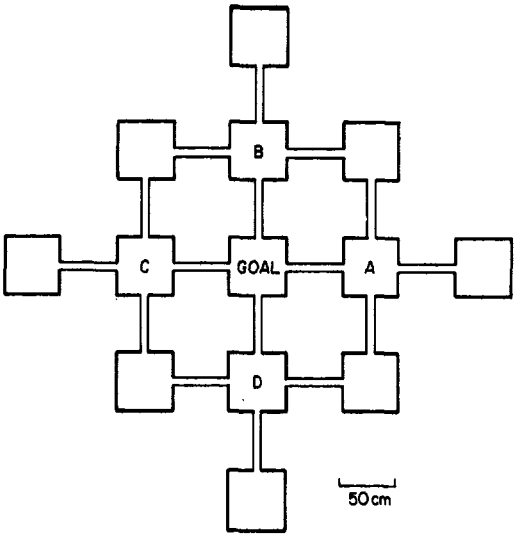


Figure 8. *Maze used to test the ability of rats to use a shortcut*—The rats were first trained to go from the three start locations (A, B, and C) to the center goal. They were then moved to the fourth start (D) and were rewarded for choosing the same goal.

independent discriminations (one from each start box) but had learned a general principle (go to the center goal) they could apply from a brand-new start location.

Rats in a second group were taught to move in a particular direction, for example, toward the top of the diagram in Figure 8. All other aspects of the testing procedure were the same. When given a transfer test, these animals performed according to their rule, choosing the arm that led in the correct direction. Consequently, correct performance on the transfer test was not due to some characteristic of the test itself but to the experience the rats had prior to it.

A more recent examination of the characteristics of maps and their distinction from routes has been offered by O'Keefe and Nadel (1978, p. 100) and is summarized in Table 1. As is apparent, their notion of a map also emphasizes behavioral flexibility and the ability of a mapper to arrive at novel (never before tried) solutions to an old problem, which should have the "ah-ha" characteristics of "insight" (Köhler, 1925).

The successful use of maplike structures has been well demonstrated in chimpanzees by the work of Menzel (1973, 1978). In one case, chimps were carried by a handler around a large compound and watched while the experimenter hid food in 18 different places. The path taken by the experimenter and the handler was tortuous and often crossed over previous paths. After all the food was hidden, the chimp was returned to a holding cage and was then released to hunt for the food. There were two questions: First, could the chimp remember where the experimenter had placed food? Second, could the chimp be efficient in obtaining the food, picking it up in an orderly fashion to minimize the distance covered? The results indicated that the chimps recovered about 12 of the 18 pieces of food and did so in an efficient manner, usually taking a large circular route through the compound and rarely recrossing a previous path. These data indicate that the chimps had a good memory for the locations in which food was placed and, more importantly, that they were able to organize their search behavior in an efficient manner, preferring relatively direct, new paths to the longer ones taken by the experimenter and the handler.

The cognitive mapping experiments are important because they demonstrate abilities that generally have been ignored in psychologists' thinking about cognitive processes in animals. In order to perform correctly in a shortcut transfer

TABLE 1

Properties of the Taxon and Locale Systems

Behavior	Taxon (route)	Locale (map)
Motivation for learning	Biological need: to obtain reward or avoid punishment	Cognitive curiosity: to construct and update a map of the environment
Learning change	Incremental or decremental	All or none
Persistence	High, especially orientation hypotheses	Low
Temporal changes after activation	Marked changes in threshold and strength as a function of time after activation; sensitive to intertrial interval	Minimal changes with time after activation; insensitive to intertrial interval
Interference between similar items	High	Low

Note. From *The Hippocampus as a Cognitive Map* by John O'Keefe and Lynn Nadel, Oxford University Press, 1978. Copyright 1978 by Oxford University Press. Reprinted by permission.

procedure, an animal must be able to interpret correctly an environmental perspective that has never been seen before and must then have a behavior flexible enough to choose the appropriate response. This type of behavior is very difficult to model with associationistic theories and reflects a general lack of understanding of how it might come about. There are at least two challenges provided by the demonstration of accurate performances by animals in a shortcut transfer experiment. The first is an empirical one, focusing on the conditions in which accurate mapping occurs and on the documentation of the experimental parameters necessary for successful mapping. The second is a theoretical one, developing both theories and models to explain the behavior, especially in mammals as simple as rats.

Behavioral Ecology

Animals search for food not only in laboratory mazes but also in their natural environments. Particularly relevant here is the question of how an animal decides where to search for food and the extent to which this decision process in the natural habitat is related to that in a laboratory maze.

Ecologists describing predators searching for prey use the terms *predator* and *prey* in a very general sense (Charnov, 1976; Covich, 1976; Krebs, 1973; Krebs, Ryan, & Charnov, 1974; Reichman, 1975; Royama, 1970; Schoener, 1971).[4] A predator is any animal or plant that is about to do the eating, and a prey is any animal or plant that is about to be eaten. Discussions of foraging assume that predators search in a "patchy" environment, one in which food resources are unequally distributed so that one patch may contain a higher density of prey items than another. Thus, if all the other parameters are equal, predators searching in patches with high prey density ought to have a selective advantage over conspecifics searching in patches with low prey density, because the amount of food obtained per unit of time/energy should be greater. A critical question for any predator, then, is the choice of the patch in which to search.

Although many different predator–prey systems have been examined, of most importance to psychologists studying animal cognition are examples of *win–stay* and *win–shift strategies* (Levine, 1959). Consider a predator that has "won" on a previous foraging trip and has obtained prey. The question on the next trip is whether to "stay" and return to the original location to seek food or to "shift" and seek food elsewhere. Examples of both these strategies may be found in natural conditions, and in each case the strategy used by the predator appears to be appropriate for the distribution of prey in the environment.

A win–shift strategy is followed by nectar-feeding birds. Kamil (1978) studied the search patterns of the Hawaiian honeycreeper (*Loxops virens*), a bird that obtains nectar from flowers on the *mamane* tree. Kamil reported that after these birds visit a flower, they usually don't return to that flower for at least several hours. The birds do not use a response chain (such as starting at one end of their territory and flying north), nor do they produce some kind of change in the flower itself. Consequently, they follow the win–shift strategy by remembering which of the flowers they have recently visited and then searching elsewhere.

Examples of win–stay strategies are also found in birds. Smith (1974a, 1974b) described the search behavior of the English thrush hunting for worms and grubs in a meadow. After finding food, the bird slowed its rate of forward movement and turned more often, effectively keeping it in the same location. Similar results have been found by Zach and Falls (1976) for the ovenbird.

Both the win–shift and win–stay strategies are adapted to the characteristics of the respective predators' food supply (see also Whitham, 1977). The flowers used by the honeycreeper contain so little nectar that the bird removes all of it during a single visit. Because the flower takes time to replenish its nectar, the optimal strategy for the honeycreeper is to search elsewhere until sufficient time has passed for the flower to have a full supply of nectar again. In contrast, the worms and grubs sought by the English thrush and the ovenbird were clustered together, so finding one prey item increased the probability of finding additional items nearby.

Laboratory tests can be used to investigate search strategies if the tests evoke the same behavioral mechanisms that operate during foraging

[4] A recent symposium relating the ecological aspects of optimal foraging to the psychological aspects of discrimination was held at the meeting of the Animal Behavior Society, Seattle, June 21–23, 1978. The papers presented there can be found in Kamil and Sargent's (in press) book.

in natural conditions (Gray, in press; Gray & Tardif, in press).[5] Furthermore, the theoretical framework developed by ecologists studying foraging suggests questions and analyses that might not have been readily apparent from a more psychological viewpoint. Experiments from a variety of sources suggest that rats have a search strategy similar to that of the Hawaiian honeycreeper and tend to shift among available food sources (Barnett, 1975; Calhoun, 1963; Olton, 1972; Olton, Walker, Gage, & Johnson, 1977). If such is the case, there are two predictions for maze behavior. First, given equally rewarded alternatives, rats ought to have an initial preference for shifting among these alternatives rather than for repeating choices to one of them. Second, given a food distribution that supports either a shift or a stay strategy, rats ought to learn the shift strategy faster than the stay.

An experiment to test these predictions (Olton, Handelmann, & Walker, in press) used a version of the Maier (1932) three-table problem. Three tables were connected to a center choice area, as is illustrated in Figure 9. A food-deprived rat was placed on one of the three tables and was allowed to eat food there. The rat was picked up by the experimenter and carried to another table that functioned as the start area. The rat left the start box and was faced with a choice of returning to the table on which it was placed or going to the other table. Returning to the table on which it was placed was defined as win–stay; choosing the other table was defined as win–shift. Both the table on which the animal was placed to eat food and the table that functioned as the start box changed from trial to trial in a haphazard manner; on each trial the rat had to remember the table on which it was placed in order to respond appropriately.

The results support both the predictions made above. First, rats rewarded for choosing either table had an initial preference to shift, to go to the table on which they had not been placed. Second, rats selectively rewarded for choosing the table on which they had not been placed learned this win–shift discrimination faster than rats selectively rewarded for returning to the table on which they had been placed. These results can also be expressed in the terminology of optimal foraging theory. When food was equally distributed among all the patches, rats preferred to shift. When food was unequally distributed among the patches, rats matched their food-searching strategy to the actual food distribution but learned the win–shift strategy faster than the win–stay one (see also Olton & Schlosberg, 1978).

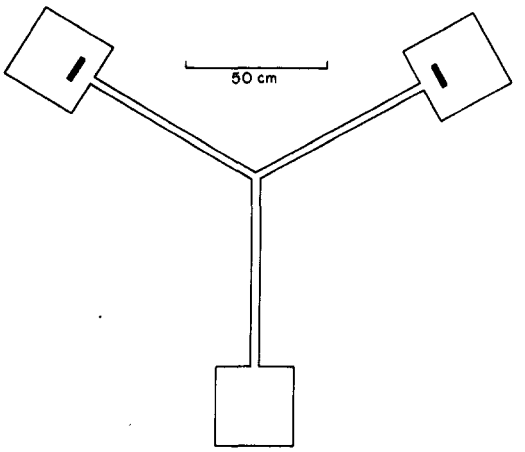


Figure 9. The elevated three-arm maze used to investigate win–shift and win–stay strategies in rats —The rats were first placed on one of the tables. They were then moved to a second table (which functioned as a start box) and were allowed to choose between returning to the table on which they had been placed and going to the remaining table. (From "Characteristics of Spatial Memory and Optimal Foraging in Rodents" by David S. Olton, G. E. Handelmann, and J. A. Walker. In A Kamil and T. Sargent (Eds.), *Optimal Foraging*, Garland/STPM, 1979. Copyright 1979 by STPM, Inc. Reprinted by permission.)

To the extent that the environment places foraging strategies under selective pressure, adaptation to that pressure ought to result in a selective advantage. There are two ways in which this adaptation might appear (Gray, in press; Gray & Tardif, in press). One is ontogenetic. The foraging behavior of each individual ought to be responsive to experience, so the distribution of food supplies in the environment during infancy might influence the predisposition to follow certain types of strategies as an adult. The other is phylogenetic. The foraging behavior of a species might be influenced by the distribution of food supplies over many generations, with genetic selection for individuals following the most appropriate strategy, resulting in the development of a species-specific behavior.

Both of these predictions can be tested in laboratory mazes, using the same general procedure

[5] This strategy has been elaborated by Kamil (Note 8, Note 9).

outlined in the three-table problem. To the extent that an individual's experience in an environment influences subsequent food-searching strategy, animals raised in environments in which food sources are consistently moving around ought to have a predisposition to follow a shift strategy, and animals raised in an environment in which there is a single, reliable food source ought to have a predisposition to follow a stay strategy. To the extent that food-searching strategies become a species-typical characteristic, species that search for changing food sources (such as nectar-feeding birds) ought to have a predisposition to follow a shift strategy in the laboratory, and species that search for reliable food sources ought to have a predisposition to follow a stay strategy. These are clear predictions and can be tested in maze tasks without difficulty.

Behavioral Flexibility Revisited

In light of the experiments just reviewed, an emphasis on behavioral stereotypy and path following in mazes is clearly inappropriate. Indeed, behavioral flexibility appears to be the rule rather than the exception. The type of apparatus in which the animal is tested may have some effect (Restle, 1957), but the major variable seems to be procedural rather than structural. First, stereotypy has been reported in both alley (Carr & Watson, 1908) and elevated (Dennis, 1932) mazes. Second, behavioral flexibility is greatest in situations in which this tendency is positively reinforced (as in Olton's experiments using the radial-arm maze) or in situations in which it is neither reinforced nor punished (as in the experiments of Dashiell and Bättig). In short, flexibility seems to be the hallmark of spatial behavior; stereotypy occurs only in situations where flexibility is punished and the most effective means of gaining reward is to repeat the same response over and over again.

The general focus on behavioral stereotypy has led to two problems in animal behavior today. First, our impression of animals' cognitive abilities has been vastly restricted: "If it is true that to solve a problem requires insight into the presented situation, then behavior *constancy militates against insight* [italics in the original]. . . . Hence it may be asserted that behavior constancy delays or prevents insight" (Hamilton & Ellis, 1933, p. 138). Because laboratory experiments have generally been designed to elicit behavioral constancy, animals have not been given the opportunity to demonstrate the insightful, flexible behaviors of which they are capable. Consequently, we have little empirical information about their abilities. Second, because the empirical data base is deficient, we lack the theoretical framework in which to put the new observations about spatial abilities: "When one considers this point, the reason for the tendency of thought in comparative psychology of the last few decades becomes at once obvious. We are too prone to draw our theoretical conclusions from the behavior which appears at a stage in the learning process which is not the significant one. For if our problems in animal learning are to be solved, we must achieve an understanding of the rationale of *achievement* [italics in original], rather than the conditions underlying mechanization" (Gingerelli, 1929, p. 260).

Conclusions

An animal should be made to do difficult things only in the line of its inherent abilities. (Ingersoll, quoted in Small, 1900, p. 133)

Many of the questions currently of interest in cognitive psychology concern the ability "to do difficult things": to remember a great deal of information, to put this information together in novel ways, and to respond adaptively to fluctuations in environmental conditions. The data reviewed in this article demonstrate that in spatial tasks, rats and other animals have a marked ability to remember information and to use this information appropriately in a flexible, adaptive manner. The data already gathered from recent maze experiments provide a challenge to current thinking in cognitive psychology, and if the progress over the past few years is any indication, maze testing offers a unique opportunity to address the issues that are of importance today. In short, animals seem to have a strong predisposition to follow spatial strategies, and by using this "inherent ability" instead of trying to suppress it, we may be able to advance our understanding of animal cognition substantially.

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